

Module 2: Eigenvalues and Eigenvectors

1 The Eigenvalue Problem

This section tackles the eigenvalue problem, a special equation involving a matrix A , a scalar λ , and a vector v . Solving it reveals how the matrix transforms vectors and identifies special vectors that only change in scale, not direction. We will crack the code of this equation and unlock the power of eigenvalues and eigenvectors!

Definition. If A is an $n \times n$ matrix, do there exist non-zero vectors $\bar{x} \in \mathbb{R}^n$ such that

A $\bar{x}$ is a scalar multiple of $\bar{x}$, i.e., $A\bar{x} = \lambda\bar{x}$? This scalar, often denoted by the Greek letter lambda, is called the eigenvalue of A , and the non-zero vector $\bar{x}$ is called the eigenvector of A corresponding to λ .

Remark. Note that we omit the case $\bar{x} = \bar{0}$ since $A\bar{0} = \lambda\bar{0}$ is true for all values of λ . An eigenvalue $\lambda = 0$, however, is possible. The equation

$$A\bar{x} = \lambda\bar{x}$$

is equivalent to

$$(A - \lambda I)\bar{x} = \bar{0}, \quad \bar{x} \neq \bar{0} \quad (*)$$

where I is the $n \times n$ identity matrix. If equation () is to have non-zero solutions, then λ must be chosen so that the $n \times n$ matrix $A - \lambda I$ is singular. That is, $\det(A - \lambda I) = 0$. Therefore the eigenvalue problem consists of two parts:*

- 1. Find all scalars λ such that the matrix $A - \lambda I$ is singular, i.e. $\det(A - \lambda I) = 0$.*
- 2. Given that $A - \lambda I$ is singular, find all the non-zero vectors $\bar{x}$ such that $(A - \lambda I)\bar{x} = \bar{0}$.*

Clearly if we know an eigenvalue of A , then the elementary row operation techniques provide an efficient way to find the eigenvectors.

2 The Spectral Decomposition

The goal of this module is to prove the Spectral Theorem, stated as follows.

2.1 Spectral Theorem

Let $A \in \mathbb{R}^{n \times n}$ be symmetric ($A^T = A$). Then there exists an orthonormal basis of eigenvectors of A , $\{v_1, \dots, v_n\}$, such that

$$A = V \Lambda V^{-1} = V \Lambda V^T$$

where $V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ and Λ is diagonal and contains the eigenvalues of A .

Let's begin with a reminder of the following fundamental concept of Linear Algebra.

Let $A \in \mathbb{R}^{n \times n}$. A scalar-vector pair $(\lambda, v) \in \mathbb{R} \times \mathbb{R}^n$ with $v \neq 0$ is an eigenvalue/eigenvector pair of A if

$$Av = \lambda v$$

Geometrically, eigenvectors are ones for which when they are multiplied by A , the product does not change direction (except for possibly a backward directional change), but rather the eigenvector is stretched in its same direction by some scalar factor. First, let's see how we can compute eigenvalue/eigenvector pairs for a given matrix. To start, we need the definition and properties of determinants.

Definition. Given 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, its determinant is

$$\det(A) := ad - bc$$

For a general $A \in \mathbb{R}^{n \times n}$, its determinant is defined iteratively as

$$\det(A) = \sum_{k=1}^n (-1)^k a_{1k} \det(M_k)$$

where the minor, M_k , is the $(n-1) \times (n-1)$ matrix formed by deleting the first row and k -th column of A .

Note that $\det(A)$ is always a scalar, and has the following important properties.

Theorem 7.3. Let $A, B \in \mathbb{R}^{n \times n}$. Then the following hold:

- (1) $\det(I) = 1$
- (2) $\det(A^T) = \det(A)$
- (3) $\det(AB) = \det(A) \det(B)$
- (4) $\det(A^{-1}) = \frac{1}{\det(A)}$ provided A is invertible
- (5) $\det(\alpha A) = \alpha^n \det(A)$ for any $\alpha \in \mathbb{R}$
- (6) If A has eigenvalues $\lambda_1, \dots, \lambda_n$, then $\det(A) = \prod_{i=1}^n \lambda_i$
- (7) $\det(A) \neq 0$ if and only if A is invertible.

Now eigenvalue/eigenvector pairs are distinctly related to determinants in the following way.

2.2 Proposition

Let $A \in \mathbb{R}^{n \times n}$. Then $(\lambda, v) \in \mathbb{R} \times \mathbb{R}^n$ with $v \neq 0$ is an eigenvalue/eigenvector pair of A if and only if

$$\det(A - \lambda I) = 0$$

2.2.1 Proof

For the forward direction, suppose that $Av = \lambda v$. Then $(A - \lambda I)v = 0$, which means that $v \in N(A - \lambda I)$. Since $v \neq 0$, $A - \lambda I$ has nontrivial nullspace, and hence is not invertible; so by Theorem 4.4(7), $\det(A - \lambda I) = 0$.

To see the converse, if $\det(A - \lambda I) = 0$, then A is not invertible, and hence has nontrivial nullspace. Thus there exists a nonzero v such that $(A - \lambda I)v = 0$, which implies that $Av = \lambda v$. Something important to note is that even if a matrix is real-valued, its eigenvalues can be complex. Indeed, if

$$A := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

then the solution to $\det(A - \lambda I) = \lambda^2 + 1 = 0$ is $\lambda = \pm i$ where i is the imaginary unit. Nonetheless, an important class of matrices do in fact have real eigenvalues as the following proposition shows.

2.3 Proposition

Let $A \in \mathbb{R}^{n \times n}$ be symmetric (i.e., $A^\top = A$). Then all eigenvalues of A are real.

2.3.1 Proof

Suppose (λ, v) is an eigenvalue/eigenvector pair of A . By conjugate-linearity in the second argument for inner products, we have

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle Av, v \rangle = \langle v, A^\top v \rangle = \langle v, Av \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle \quad (2)$$

Since $v \neq 0$, $\langle v, v \rangle \neq 0$, thus (2) implies that $\lambda = \bar{\lambda}$, hence λ must be real. Next we will note that distinct eigenvalues yield orthogonal eigenvectors.

2.4 Proposition

Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric, and (λ_1, ν_1) and (λ_2, ν_2) are eigenvalue/eigenvector pairs of $A \in \mathbb{R}^{n \times n}$. If $\lambda_1 \neq \lambda_2$, then ν_1 is orthogonal to ν_2 .

2.4.1 Proof

Note that (since $\lambda_1, \lambda_2 \in \mathbb{R}$)

$$\lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \langle Av_1, v_2 \rangle = \langle v_1, Av_2 \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$$

This implies that $(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0$, but $\lambda_1 - \lambda_2 \neq 0$ by assumption, so v_1 and v_2 must be orthogonal.

This leads to the following observation:

2.5 Proposition

(Easy case of the Spectral Theorem). Suppose that $A \in \mathbb{R}^{n \times n}$ is symmetric and has n distinct, nonzero eigenvalues, $\lambda_1, \dots, \lambda_n$. Then A has n orthonormal eigenvectors, and we may write

$$A = SAS^{-1} = S\Lambda S^\top = \sum_{i=1}^n \lambda_i S_{:i} S_{:i}^\top$$

where the columns of S are the eigenvectors of A and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Remark. Note that this result proves the Spectral Theorem in the case that A has distinct eigenvalues, so the only case we have left to consider is that when A has repeated eigenvalues. We will illustrate a couple of ways one can finish the proof of the general case.

Our first argument for proving the Spectral Theorem is a more constructive one which boils down to repeatedly doing Gram-Schmidt on blocks of the initial matrix. Suppose that (λ_1, v_1) is any eigenvalue/eigenvector pair of A and note that without loss of generality we may assume that v_1 is a unit vector (if not divide by its norm). We know that there is an orthonormal basis for $\mathbb{R}^n$ containing v_1 , say $\{v_1, w_2, \dots, w_n\}$. Let Q_1 be the matrix with these basis vectors as columns. Then by orthonormality of its columns, we have

$$Q_1 A Q_1^\top = \begin{bmatrix} \lambda_1 & \mathbf{0} \\ \mathbf{0} & B_1 \end{bmatrix}$$

for some symmetric matrix $B_1 \in \mathbb{R}^{(n-1) \times (n-1)}$.

Before proceeding we need the following crucial observation

2.6 Lemma

Let $A \in \mathbb{R}^{n \times n}$. If $Q \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then $Q A Q^\top$ has the same eigenvalues as A .

2.6.1 Proof

Let (λ, v) be an eigenvalue/eigenvector pair of $QAQ^\top$. Then $QAQ^\top v = \lambda v$, and by multiplying by $Q^\top$ on the left, we see that

$$AQ^\top v = \lambda Q^\top v$$

which means that $(\lambda, Q^\top v)$ is an eigenvalue/eigenvector pair of A . Since both A and $Q_A^\top$ have the same rank and this argument works for all eigenpairs, the claim is proven.

Remark. Now we may apply the conclusion of the lemma to note that $Q_1 A Q_1^\top$ has the same eigenvalues as A , hence the eigenvalues of B_1 are $\lambda_2, \dots, \lambda_n$. Now we iterate the same argument as above to find an orthogonal $\tilde{Q}$ for which

$$\tilde{Q} B_1 \tilde{Q}^\top = \begin{bmatrix} \lambda_2 & 0 \\ 0 & B_2 \end{bmatrix},$$

and we set

$$Q_2 := \begin{bmatrix} 1 & 0 \\ 0 & \tilde{Q} \end{bmatrix}$$

Now we have that

$$Q_2 Q_1 A Q_1^\top Q_2^\top = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & B_2 \end{bmatrix}$$

Iterating this procedure, we find $Q_1, \dots, Q_n$ which diagonalize A into $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ as desired, and we take $S := Q_n \dots Q_1$ to get the desired decomposition.

2.6.2 Second Proof

Our second argument is nonconstructive, but follows from somewhat more basic Linear Algebra facts. We begin with the following observation.

2.7 Lemma

Let $A \in \mathbb{R}^{n \times n}$ be symmetric, and let S be any subspace of $\mathbb{R}^n$. If $Ax \in S$ for every $x \in S$, then $Ay \in S^\perp$ for every $y \in S^\perp$.

2.7.1 Proof

By assumption, $\langle Ax, y \rangle = 0$, but by symmetry, $\langle Ax, y \rangle = \langle x, Ay \rangle$, so $Ay \in S^\perp$ since x is an arbitrary element of S .

Lemma 7.12. If $A \in \mathbb{R}^{n \times n}$ is symmetric and S is a nonzero subspace of $\mathbb{R}^n$ such that $Ax \in S$ for every $x \in S$, then S contains an eigenvector of A .

Proof

First of all, we know that S contains an orthonormal basis by Theorem 3.4; call it $\{u_1, \dots, u_k\}$. Since $Au_j \in S$ for all j , there are scalars $\{r_{ij}\}_{i,j=1}^k$ such that

$$Au_j = \sum_{i=1}^k r_{ij} u_i$$

Indeed $r_{ij} = \langle Au_j, u_i \rangle$. Let R be the $k \times k$ matrix with entries $R_{ij} = r_{ij}$. Then note that R is symmetric because $r_{ji} = \langle Au_i, u_j \rangle = \langle u_i, Au_j \rangle = \langle Au_j, u_i \rangle = r_{ij}$.

Now suppose that (λ, v) is an eigenvalue/eigenvector pair of R . Then define $w := \sum_{j=1}^k v_j u_j$ where v_j is the j -th component of the vector v , and u_j is still one of the basis vectors for S . Then we have

$$\begin{aligned} Aw &= \sum_{j=1}^k v_j Au_j \\ &= \sum_{j=1}^k v_j \sum_{i=1}^k r_{ij} u_i \\ &= \sum_{i=1}^k \left(\sum_{j=1}^k r_{ij} v_j \right) u_i \\ &= \sum_{i=1}^k (Rv)_i u_i \\ &= \sum_{i=1}^k \lambda v_i u_i \\ &= \lambda w \end{aligned}$$

Hence (λ, w) is an eigenvalue/eigenvector pair of A .

Now we are ready to complete the proof of the Spectral Theorem.

Proof of the Spectral Theorem. Let (λ_1, ν_1) be an eigenvalue/eigenvector pair of A , and set $S_1 := \text{span}(\nu_1)$. Note that $A(\alpha \nu_1) = \lambda \alpha \nu_1 \in S_1$, so by Lemma 4.10, $Ay \in S_1^\perp$ for every $y \in S_1^\perp$. If $S_1^\perp \neq \{0\}$, then Lemma 4.11 implies that there exists an eigenvector $\nu_2 \in S_1^\perp$ (which is necessarily orthogonal to ν_1). Now if $n > 2$, then set $S_2 := \text{span}(\nu_1, \nu_2)$, and note that $Ax \in S_2$ for every $x \in S_2$ by a similar argument to the first case. Again apply Lemmas 4.10 and 4.11 to get an orthogonal eigenvector $\nu_3 \in S_2^\perp$. Continue in this manner to choose $\nu_1, \dots, \nu_n$ and the proof is complete.

Now let us note that the Spectral Theorem gives us a representation of a matrix in terms of the sum of rank 1 matrices.

Corollary Let $A \in \mathbb{R}^{n \times n}$ be symmetric, and let $\{v_i\}_{i=1}^n$ be an orthonormal basis of eigenvectors of A with corresponding eigenvalues $\{\lambda_i\}_{i=1}^n$. Then

$$A = \sum_{i=1}^n \lambda_i v_i v_i^\top$$

Proof. First note that

$$\begin{aligned} V\Lambda &= \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \begin{bmatrix} \lambda_1 & \mathbf{0} & \cdots & 0 \\ \mathbf{0} & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \mathbf{0} & \cdots & 0 & \lambda_n \end{bmatrix} \\ &= \begin{bmatrix} \lambda v_1 & \cdots & \lambda_n v_n \end{bmatrix}. \end{aligned}$$

Since $V^{-1} = V^\top$, we have

$$V\Lambda v^{-1} = \begin{bmatrix} \lambda v_1 & \cdots & \lambda_n v_n \end{bmatrix} \begin{bmatrix} v_1^\top \\ \vdots \\ v_n^\top \end{bmatrix} = \sum_{i=1}^n \lambda_i v_i v_i^\top$$

as required.

3 Python Programming Tasks

3.1 Learning Tasks for Spectral Decomposition (Eigendecomposition) with Python

These tasks will guide learners through the fundamental concepts of spectral decomposition, hands-on implementation in Python, and the exploration of applications using visualizations. Learners will gain an understanding of eigenvalues, eigenvectors, matrix reconstruction, and how these are applied in various areas of linear algebra and data science.

3.2 Task 1: Review the Theory of Spectral Decomposition

Goal: Ensure a clear understanding of the theoretical basis of spectral decomposition, including eigenvalues and eigenvectors.

3.2.1 Instructions:

1. Define spectral decomposition and explain the role of eigenvalues and eigenvectors in this process.
2. Show that for any matrix A , spectral decomposition takes the form $A = QQ^\top$, where Q contains eigenvectors and Λ is a diagonal matrix of eigenvalues.
3. Derive the spectral decomposition for the following matrix by hand:

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

4. Compute the eigenvalues and eigenvectors manually and verify the reconstruction of the matrix.

Expected Outcome: A written explanation and manual calculations showing the spectral decomposition of a simple 2×2 matrix.

3.3 Task 2: Implement Spectral Decomposition in Python

Goal: Write Python code to perform spectral decomposition of a square matrix and reconstruct the matrix using eigenvalues and eigenvectors.

3.3.1 Instructions:

1. Implement a Python function that:
 - Takes a square matrix A as input.
 - Computes the eigenvalues and eigenvectors using NumPy's `np.linalg.eig()` function.
 - Reconstructs the matrix using the formula $A = Q\Lambda Q^{-1}$.
2. Test your implementation on the following matrix:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

3. Verify that the reconstructed matrix matches the original matrix.

Expected Outcome: A working Python function that outputs the eigenvalues, eigenvectors, and the reconstructed matrix, verifying the correctness of the decomposition.

3.4 Task 3: Explore Visualization of Eigenvectors

Goal: Use Python's Matplotlib to visualize eigenvectors as vectors in 2D space.

3.4.1 Instructions:

1. Write Python code to:
 - Plot the original matrix A as a heatmap.
 - Plot the eigenvectors of the matrix as arrows (vectors) on a 2D plane.

- Scale the length of each eigenvector according to the corresponding eigenvalue.
2. For the matrix $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, visualize the eigenvectors and label them with their corresponding eigenvalues.
Expected Outcome: A visualization with two arrows representing the eigenvectors of matrix A, scaled by their eigenvalues, alongside a heatmap of the matrix.

3.5 Task 4: Spectral Decomposition in 3D

Goal: Extend your Python implementation to handle 3D matrices and visualize eigenvectors in 3D space.

3.5.1 Instructions:

1. Modify your spectral decomposition function to handle 3×3 matrices.
2. Use a 3×3 matrix such as:

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 5 \end{bmatrix}$$

3. Write Python code to visualize the eigenvectors of this matrix in 3D using `mpl_toolkits.mplot3d`.
4. Display the eigenvectors as arrows originating from the origin and scaled by their eigenvalues.
Expected Outcome: A 3D plot showing the eigenvectors of the 3×3 matrix, scaled by their corresponding eigenvalues.

3.6 Task 5: Numerical Stability and Symmetric Matrices

Goal: Investigate the behavior of spectral decomposition for symmetric matrices and discuss the numerical stability of the process.

3.6.1 Instructions:

1. Explain why the eigenvalues of symmetric matrices are always real and why the eigenvectors are orthogonal.
2. Test your spectral decomposition code with the following symmetric matrix:

$$A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

3. Verify the orthogonality of the eigenvectors by checking that their dot products are zero.
4. Discuss the numerical stability of eigendecomposition when dealing with symmetric matrices, and compare it with the decomposition of a non-symmetric matrix.

Expected Outcome: A report explaining the properties of symmetric matrices, eigenvalue behavior, and numerical stability, along with verification of orthogonality in your Python code.

3.7 Task 6: Apply Spectral Decomposition to Dimensionality Reduction

Goal: Understand how spectral decomposition can be applied to dimensionality reduction techniques such as Principal Component Analysis (PCA).

3.7.1 Instructions:

1. Research how PCA uses the eigenvalues and eigenvectors of the covariance matrix to perform dimensionality reduction.
2. Download a standard dataset, such as the Iris dataset, and:
 - Compute the covariance matrix of the dataset.
 - Perform spectral decomposition of the covariance matrix.
 - Use the eigenvectors to project the data onto the principal components (reduce the dimensionality to 2D).
3. Visualize the original dataset and the reduced dataset in 2D.

Expected Outcome: A 2D plot of the original and reduced datasets, along with a clear explanation of how PCA is related to spectral decomposition.

3.8 Task 7: Solve Practical Problems Using Spectral Decomposition

Goal: Apply spectral decomposition to solve practical problems such as solving linear systems and computing matrix powers.

3.8.1 Instructions:

1. Use the following matrix A :

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

2. Compute its spectral decomposition.

3. Use the decomposition to:

- Solve the linear system $Ax = b$ for $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
 - Compute the matrix power A^5 using the spectral decomposition.
4. Compare the result with the direct computation of A^5 using `np.linalg.matrix_power()`.

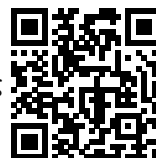
Expected Outcome: A step-by-step solution of the linear system and the matrix power computation using both spectral decomposition and direct methods, verifying the results.

4 Content Curated Video

Visual learners may benefit from the following YouTube videos that explain the concepts of eigenvalues, eigenvectors, and the Spectral Theorem. These videos offer step-by-step explanations and visualizations to help solidify your understanding of spectral decomposition.

- **Video** Visit the URL below to view a video:

<https://www.youtube.com/embed/PFDu9oVAE-g>



Eigenvalues and Eigenvectors — Explained by 3Blue1Brown

4.1 Python Code for Eigenvalue Calculation

To compute eigenvalues and eigenvectors for a given matrix, we can use Python's NumPy library. Here's an example code to calculate eigenvalues and eigenvectors:

4.1.1 Eigenvalue and Eigenvector Calculation in Python

```
import numpy as np
\# Define a symmetric matrix
A = np.array ([[4, 1],
               [1,3]])

\# Compute eigenvalues and eigenvectors}
eigenvalues, eigenvectors = np.linalg.eig (A)
print("Eigenvalues:", eigenvalues)
print("Eigenvectors:\textbackslash n", eigenvectors)
```

This code computes the eigenvalues and eigenvectors of a symmetric matrix A , which is critical for understanding the spectral decomposition process. For more examples and code resources, visit the following links:

- <https://github.com/topics/eigen-decomposition>: Eigenvalue Decomposition Code - GitHub

This code computes the eigenvalues and eigenvectors of a symmetric matrix A , which is critical for understanding the spectral decomposition process.

5 Recommended Supplementary Reading

For a deeper understanding of the Spectral Theorem and linear algebra concepts related to eigenvalues and eigenvectors, refer to the following textbooks:

1. Textbook: Read Chapter 5 ("Eigenvalues and Eigenvectors") from "Introduction to Linear Algebra" by Gilbert Strang.
2. Online Article: <https://www.mathsisfun.com/algebra/eigenvalue.html> Eigenvalues and Eigenvectors Explained Visually

These resources provide comprehensive coverage of matrix theory, spectral decomposition, and various applications of eigenvalues in real-world problems. These resources provide comprehensive coverage of matrix theory, spectral decomposition, and various applications of eigenvalues in real-world problems.